In [43]:
```
import numpy as np
import pandas as pd
```

In [31]:
```
df = pd.read_excel("marks.xlsx", sheet_name="Sheet1", header=0, index_col=None)
```

In [51]:
```
df
```

Out[51]:

	Roll No	Name	Marks
0	101	Jay	25
1	102	Pari	23
2	103	Riya	27
3	104	Megha	26
4	105	Shreya	21

In [43]:
```
df.loc[2,"Marks"] = 28
```

In [49]:
```
df["Marks"][2] = 27
```

```
C:\Users\mania\AppData\Local\Temp\ipykernel_2776\2451753961.py:1: FutureWarning:
ChainedAssignmentError: behaviour will change in pandas 3.0!
You are setting values through chained assignment. Currently this works in certai
n cases, but when using Copy-on-Write (which will become the default behaviour in
pandas 3.0) this will never work to update the original DataFrame or Series, beca
use the intermediate object on which we are setting values will behave as a copy.
A typical example is when you are setting values in a column of a DataFrame, lik
e:

df["col"][row_indexer] = value

Use `df.loc[row_indexer, "col"] = values` instead, to perform the assignment in a
single step and ensure this keeps updating the original `df`.

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stabl
e/user_guide/indexing.html#returning-a-view-versus-a-copy

  df["Marks"][2] = 27
C:\Users\mania\AppData\Local\Temp\ipykernel_2776\2451753961.py:1: SettingWithCopy
Warning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stabl
e/user_guide/indexing.html#returning-a-view-versus-a-copy
  df["Marks"][2] = 27
```

In [45]:
```
df
```

Out[45]:

	Roll No	Name	Marks
0	101	Jay	25
1	102	Pari	23
2	103	Riya	28
3	104	Megha	26
4	105	Shreya	21

In [35]:
```
df['Name']
```

Out[35]:
```
0       Jay
1      Pari
2      Riya
3     Megha
4    Shreya
Name: Name, dtype: object
```

In [39]:
```
df['Marks'].mean()
```

Out[39]:
```
np.float64(24.4)
```

In [53]:
```
l1 = [1,2,3]
```

In [55]:
```
l2 = l1
```

In [57]:
```
l1.append(4)
```

In [59]:
```
l1
```

Out[59]:
```
[1, 2, 3, 4]
```

In [61]:
```
l2
```

Out[61]:
```
[1, 2, 3, 4]
```

In [63]:
```
l3 = l1.copy()
```

In [65]:
```
l1.append(5)
```

In [67]:
```
l1
```

Out[67]:
```
[1, 2, 3, 4, 5]
```

In [69]:
```
l3
```

Out[69]:
```
[1, 2, 3, 4]
```

In [71]:
```
df
```

Out[71]:

	Roll No	Name	Marks
0	101	Jay	25
1	102	Pari	23
2	103	Riya	27
3	104	Megha	26
4	105	Shreya	21

In [73]:
```
df.to_csv("new.csv")
```

In [1]:
```
import numpy as np
import pandas as pd
```

In [3]:
```
data = {"RollNo" : range(101,106),
        "Name" : ["Jay","Mayank","Riya","Seema","Karan"],
        "Marks" : np.random.randint(75,101,5)}
```

In [9]:
```
print(type(data))
```

```
<class 'dict'>
```

In [17]:
```
df = pd.DataFrame(data)
```

In [19]:
```
df
```

Out[19]:

	RollNo	Name	Marks
0	101	Jay	99
1	102	Mayank	91
2	103	Riya	88
3	104	Seema	95
4	105	Karan	82

In [ ]:
```
set_index() and reset_index()
```

In [21]:
```
df
```

Out[21]:

	RollNo	Name	Marks
0	101	Jay	99
1	102	Mayank	91
2	103	Riya	88
3	104	Seema	95
4	105	Karan	82

In [27]:
```
df.set_index("RollNo", inplace = True)
```

In [29]:
```
df
```

Out[29]:

RollNo	Name	Marks
101	Jay	99
102	Mayank	91
103	Riya	88
104	Seema	95
105	Karan	82

In [31]:
```
df.loc[102]
```

Out[31]:
```
Name      Mayank
Marks         91
Name: 102, dtype: object
```

In [33]:
```
df
```

Out[33]:

RollNo	Name	Marks
101	Jay	99
102	Mayank	91
103	Riya	88
104	Seema	95
105	Karan	82

In [37]:
```
df = df.reset_index()
```

In [39]:
```
df
```

Out[39]:

	RollNo	Name	Marks
0	101	Jay	99
1	102	Mayank	91
2	103	Riya	88
3	104	Seema	95
4	105	Karan	82

In [43]:
```
df = df.set_index("Name")
```

In [45]:
```
df
```

Out[45]:

Name	RollNo	Marks
Jay	101	99
Mayank	102	91
Riya	103	88
Seema	104	95
Karan	105	82

In [47]:
```
df.reset_index(drop= True, inplace = True)
```

In [109…
```
d1 = {"Department" : ["CE","CE","CE","IT","IT","IT"],
     "Semester" : [4,4,5,4,4,5],
     "Name" : ["Jay","Mayank","Riya","Seema","Karan","Neha"],
     "Marks" : np.random.randint(45,101,6)}
```

In [111…
```
df = pd.DataFrame(d1)
```

In [113…
```
df
```

Out[113…

	Department	Semester	Name	Marks
0	CE	4	Jay	50
1	CE	4	Mayank	72
2	CE	5	Riya	49
3	IT	4	Seema	70
4	IT	4	Karan	75
5	IT	5	Neha	87

In [115…
```
df = df.set_index(["Department","Semester"])
```

In [117…
```
df
```

Out[117…

		Name	Marks
Department	**Semester**		
CE	**4**	Jay	50
	4	Mayank	72
	5	Riya	49
IT	**4**	Seema	70
	4	Karan	75
	5	Neha	87

In [119…
```
df.index
```

Out[119…
```
MultiIndex([('CE', 4),
            ('CE', 4),
            ('CE', 5),
            ('IT', 4),
            ('IT', 4),
            ('IT', 5)],
           names=['Department', 'Semester'])
```

In [123…
```
df.loc["IT"]
```

Out[123…

Semester	Name	Marks
4	Seema	70
4	Karan	75
5	Neha	87

In [125…
```
df.loc[("CE",4)]
```

Out[125…

Department	Semester	Name	Marks
CE	4	Jay	50
	4	Mayank	72

In [127…
```
df
```

Out[127…

Department	Semester	Name	Marks
CE	4	Jay	50
	4	Mayank	72
	5	Riya	49
IT	4	Seema	70
	4	Karan	75
	5	Neha	87

In [133…
```
df.loc["CE"].loc[5]
```

Out[133…
```
Name     Riya
Marks      49
Name: 5, dtype: object
```

In [155…
```
df1 = pd.read_excel("C:\\Users\\mania\\OneDrive\\Desktop\\nptel course.xlsx")
```

In [157…
```
df1.head(3)
```

Out[157...

	Course ID	Discipline	Course Name	SME Name	Institute	Coordinating Institute	Duration	Typ
0	noc26-cs104	Computer Science and Engineering	Generative AI for Computer Vision	Prof. Arijit Sur	IIT Guwahati	IIT Guwahati	12 Weeks	Ne
1	noc26-cs119	Computer Science and Engineering	Introduction to Machine Learning - IITKGP	Prof. Sudeshna Sarkar	IIT Kharagpur	IIT Kharagpur	8 Weeks	Reru
2	noc26-cs133	Computer Science and Engineering	Introduction to Machine Learning	Prof. Balaraman Ravindran	IIT Madras	IIT Madras	12 Weeks	Reru

In [179...
```
arr= df1["SME Name"].unique()
```

In [183...
```
name = arr.tolist()
```

In [189...
```
for i in name:
    print(i)
```

```
Prof. Arijit Sur
Prof. Sudeshna Sarkar
Prof. Balaraman Ravindran
Prof. Sudarshan Iyengar
Prof. Tanmoy Chakraborty,
Prof. Soumen Chakraborti
Prof. Pawan Goyal
Prof. Sriram Ganapathy,
Prof. Ashwini Kodipalli,
Prof. Baishali Garai
nan
IIT Hyderabad
```

In [191...
```
df1["Click here to Join the course"].nunique()
```

Out[191... 8

In [193...
```
df1["Click here to Join the course"].value_counts()
```

Out[193...
```
Click here to Join the course
https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs104    1
https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs119    1
https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs133    1
https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs138    1
https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs171    1
https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs180    1
https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs181    1
https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs95     1
Name: count, dtype: int64
```

In [197...
```
print(np.nan)
```

```
nan
```

In [143…
```
df2 = pd.read_csv("multi_index.csv")
```

In [145…
```
df2
```

Out[145…

	Course	City	Name	Marks
0	BCA	Ahmedabad	Rahul	85
1	BCA	Ahmedabad	Anjali	95
2	BCA	Rajkot	Vivek	73
3	BCA	Rajkot	Amit	78
4	BBA	Ahmedabad	Priya	91
5	BBA	Surat	Neha	88
6	BBA	Morbi	Pooja	84
7	MCA	Rajkot	Rohan	69
8	MCA	Vadodara	Karan	76
9	MCA	Vadodara	Meera	89

In [147…
```
df2 = df2.set_index(["Course","City"])
```

In [153…
```
df2
```

Out[153…

		Name	Marks
Course	**City**		
BCA	**Ahmedabad**	Rahul	85
	Ahmedabad	Anjali	95
	Rajkot	Vivek	73
	Rajkot	Amit	78
BBA	**Ahmedabad**	Priya	91
	Surat	Neha	88
	Morbi	Pooja	84
MCA	**Rajkot**	Rohan	69
	Vadodara	Karan	76
	Vadodara	Meera	89

In [3]:
```
df = pd.read_csv("multi_index.csv")
```

In [5]:
```
df
```

Out[5]:

	Course	City	Name	Marks
0	BCA	Ahmedabad	Rahul	85
1	BCA	Ahmedabad	Anjali	95
2	BCA	Rajkot	Vivek	73
3	BCA	Rajkot	Amit	78
4	BBA	Ahmedabad	Priya	91
5	BBA	Surat	Neha	88
6	BBA	Morbi	Pooja	84
7	MCA	Rajkot	Rohan	69
8	MCA	Vadodara	Karan	76
9	MCA	Vadodara	Meera	89

In [9]:
```
df = df.set_index(["Course","City"])
```

In [11]:
```
df
```

Out[11]:

		Name	Marks
Course	**City**		
BCA	**Ahmedabad**	Rahul	85
	Ahmedabad	Anjali	95
	Rajkot	Vivek	73
	Rajkot	Amit	78
BBA	**Ahmedabad**	Priya	91
	Surat	Neha	88
	Morbi	Pooja	84
MCA	**Rajkot**	Rohan	69
	Vadodara	Karan	76
	Vadodara	Meera	89

In [41]:
```
d1 = {"Department" : ["CE","CE","CE","IT","IT","IT"],
     "Semester" : [4,4,5,4,4,5],
     "Name" : ["Jay","Mayank","Riya","Seema","Karan","Neha"],
     "Marks" : np.random.randint(45,101,6)}
```

In [45]:
```
df = pd.DataFrame(d1)
```

In [47]:
```
df
```

Out[47]:

	Department	Semester	Name	Marks
0	CE	4	Jay	91
1	CE	4	Mayank	47
2	CE	5	Riya	66
3	IT	4	Seema	95
4	IT	4	Karan	61
5	IT	5	Neha	86

In [49]:
```
df = df.set_index(["Department","Semester"])
```

In [51]:
```
df
```

Out[51]:

		Name	Marks
Department	**Semester**		
CE	**4**	Jay	91
	4	Mayank	47
	5	Riya	66
IT	**4**	Seema	95
	4	Karan	61
	5	Neha	86

In [53]:
```
df.loc["CE"]
```

Out[53]:

	Name	Marks
Semester		
4	Jay	91
4	Mayank	47
5	Riya	66

In [69]:
```
df_4 = df.xs(4, level = 1, drop_level=False)
```

Out[69]:

		Name	Marks
Department	**Semester**		
CE	**4**	Jay	91
	4	Mayank	47
IT	**4**	Seema	95
	4	Karan	61

unique(), nunique(), value_counts()

In [72]:
```
df
```

Out[72]:

		Name	Marks
Department	**Semester**		
CE	**4**	Jay	91
	4	Mayank	47
	5	Riya	66
IT	**4**	Seema	95
	4	Karan	61
	5	Neha	86

In [74]:
```
d1
```

Out[74]:
```
{'Department': ['CE', 'CE', 'CE', 'IT', 'IT', 'IT'],
 'Semester': [4, 4, 5, 4, 4, 5],
 'Name': ['Jay', 'Mayank', 'Riya', 'Seema', 'Karan', 'Neha'],
 'Marks': array([91, 47, 66, 95, 61, 86], dtype=int32)}
```

In [76]:
```
df = pd.DataFrame(d1)
```

In [78]:
```
df
```

Out[78]:

	Department	Semester	Name	Marks
0	CE	4	Jay	91
1	CE	4	Mayank	47
2	CE	5	Riya	66
3	IT	4	Seema	95
4	IT	4	Karan	61
5	IT	5	Neha	86

In [80]:
```
df["Department"].nunique()
```

Out[80]:
```
2
```

In [82]:
```
df["Department"].unique()
```

Out[82]:
```
array(['CE', 'IT'], dtype=object)
```

In [88]:
```
df["Department"].value_counts(normalize = True)*100
```

Out[88]:
```
Department
CE    50.0
IT    50.0
Name: proportion, dtype: float64
```

In [90]:
```
df
```

Out[90]:

	Department	Semester	Name	Marks
0	CE	4	Jay	91
1	CE	4	Mayank	47
2	CE	5	Riya	66
3	IT	4	Seema	95
4	IT	4	Karan	61
5	IT	5	Neha	86

In [92]:
```
df["Semester"].value_counts()
```

Out[92]:
```
Semester
4    4
5    2
Name: count, dtype: int64
```

Handling Missing Values

In [97]:
```
a1 = pd.Series([10,20,np.nan,30,40,50])
```

In [99]:
```
a1
```

Out[99]:
```
0    10.0
1    20.0
2     NaN
3    30.0
4    40.0
5    50.0
dtype: float64
```

In [105…
```
a1.nunique(dropna = False)
```

Out[105…
```
6
```

In [107…
```
a1.value_counts(dropna = False)
```

Out[107…
```
10.0    1
20.0    1
NaN     1
30.0    1
40.0    1
50.0    1
Name: count, dtype: int64
```

In [115…
```
data = {"Name" : ["Ajay","Neha","Jatin", "Sneha"],
        "Age" : [21, None, 22, 23],
```

```
        "Marks" : [45, 42, np.nan, 40],
        "City" : ["Rajkot","Morbi",None,"Ahmedabad"]}
```

In [117…
```
df = pd.DataFrame(data)
```

In [119…
```
df
```

Out[119…

	Name	Age	Marks	City
0	Ajay	21.0	45.0	Rajkot
1	Neha	NaN	42.0	Morbi
2	Jatin	22.0	NaN	None
3	Sneha	23.0	40.0	Ahmedabad

In [125…
```
df.isnull().sum().sum()
```

Out[125…
```
np.int64(3)
```

In [127…
```
df.isna()
```

Out[127…

	Name	Age	Marks	City
0	False	False	False	False
1	False	True	False	False
2	False	False	True	True
3	False	False	False	False

In [131…
```
df["Name"].isnull().sum()
```

Out[131…
```
np.int64(0)
```

In [ ]:
```
dropna() -> removes
```

In [133…
```
df
```

Out[133…

	Name	Age	Marks	City
0	Ajay	21.0	45.0	Rajkot
1	Neha	NaN	42.0	Morbi
2	Jatin	22.0	NaN	None
3	Sneha	23.0	40.0	Ahmedabad

In [139…
```
df.dropna(axis = 1, inplace = False)
```

Out[139…

	Name
0	Ajay
1	Neha
2	Jatin
3	Sneha

In [141…
```
df
```

Out[141…

	Name	Age	Marks	City
0	Ajay	21.0	45.0	Rajkot
1	Neha	NaN	42.0	Morbi
2	Jatin	22.0	NaN	None
3	Sneha	23.0	40.0	Ahmedabad

In [145…
```
df.dropna(axis = 0, how = "any")
```

Out[145…

	Name	Age	Marks	City
0	Ajay	21.0	45.0	Rajkot
3	Sneha	23.0	40.0	Ahmedabad

In [147…
```
df
```

Out[147…

	Name	Age	Marks	City
0	Ajay	21.0	45.0	Rajkot
1	Neha	NaN	42.0	Morbi
2	Jatin	22.0	NaN	None
3	Sneha	23.0	40.0	Ahmedabad

In [151…
```
df.dropna(subset = ["Marks","Age"])
```

Out[151…

	Name	Age	Marks	City
0	Ajay	21.0	45.0	Rajkot
3	Sneha	23.0	40.0	Ahmedabad

In [1]:
```
import numpy as np
import pandas as pd
```

In [3]:
```
data = {"Name" : ["Ajay","Neha","Jatin", "Sneha"],
        "Age" : [21, None, 22, 23],
        "Marks" : [45, 42, np.nan, 40],
        "City" : ["Rajkot","Morbi",None,"Ahmedabad"]}
```

In [5]:
```
df = pd.DataFrame(data)
```

In [7]:
```
df
```

Out[7]:

	Name	Age	Marks	City
0	Ajay	21.0	45.0	Rajkot
1	Neha	NaN	42.0	Morbi
2	Jatin	22.0	NaN	None
3	Sneha	23.0	40.0	Ahmedabad

In [9]:
```
df
```

Out[9]:

	Name	Age	Marks	City
0	Ajay	21.0	45.0	Rajkot
1	Neha	NaN	42.0	Morbi
2	Jatin	22.0	NaN	None
3	Sneha	23.0	40.0	Ahmedabad

In [11]:
```
df.fillna(0)
```

Out[11]:

	Name	Age	Marks	City
0	Ajay	21.0	45.0	Rajkot
1	Neha	0.0	42.0	Morbi
2	Jatin	22.0	0.0	0
3	Sneha	23.0	40.0	Ahmedabad

In [13]:
```
df
```

Out[13]:

	Name	Age	Marks	City
0	Ajay	21.0	45.0	Rajkot
1	Neha	NaN	42.0	Morbi
2	Jatin	22.0	NaN	None
3	Sneha	23.0	40.0	Ahmedabad

In [23]:
```
df["Age"] = df["Age"].fillna(df["Age"].mean())
```

In [25]:
```
df
```

Out[25]:

	Name	Age	Marks	City
0	Ajay	21.0	45.0	Rajkot
1	Neha	22.0	42.0	Morbi
2	Jatin	22.0	NaN	None
3	Sneha	23.0	40.0	Ahmedabad

In [27]:
```
df["City"] = df["City"].fillna(df["City"].mode())
```

In [29]:
```
df
```

Out[29]:

	Name	Age	Marks	City
0	Ajay	21.0	45.0	Rajkot
1	Neha	22.0	42.0	Morbi
2	Jatin	22.0	NaN	Rajkot
3	Sneha	23.0	40.0	Ahmedabad

In [31]:
```
df["Marks"] = df["Marks"].fillna(df["Marks"].min())
```

In [33]:
```
df
```

Out[33]:

	Name	Age	Marks	City
0	Ajay	21.0	45.0	Rajkot
1	Neha	22.0	42.0	Morbi
2	Jatin	22.0	40.0	Rajkot
3	Sneha	23.0	40.0	Ahmedabad

In [35]:
```
data
```

Out[35]:
```
{'Name': ['Ajay', 'Neha', 'Jatin', 'Sneha'],
 'Age': [21, None, 22, 23],
 'Marks': [45, 42, nan, 40],
 'City': ['Rajkot', 'Morbi', None, 'Ahmedabad']}
```

In [37]:
```
df = pd.DataFrame(data)
```

In [39]:
```
df
```

Out[39]:

	Name	Age	Marks	City
0	Ajay	21.0	45.0	Rajkot
1	Neha	NaN	42.0	Morbi
2	Jatin	22.0	NaN	None
3	Sneha	23.0	40.0	Ahmedabad

In [43]:
```
df.fillna(method = 'bfill')
```

```
C:\Users\mania\AppData\Local\Temp\ipykernel_19776\3673297803.py:1: FutureWarning:
DataFrame.fillna with 'method' is deprecated and will raise in a future version.
Use obj.ffill() or obj.bfill() instead.
  df.fillna(method = 'bfill')
```

Out[43]:

	Name	Age	Marks	City
0	Ajay	21.0	45.0	Rajkot
1	Neha	22.0	42.0	Morbi
2	Jatin	22.0	40.0	Ahmedabad
3	Sneha	23.0	40.0	Ahmedabad

In [45]:
```
df.ffill()
```

Out[45]:

	Name	Age	Marks	City
0	Ajay	21.0	45.0	Rajkot
1	Neha	21.0	42.0	Morbi
2	Jatin	22.0	42.0	Morbi
3	Sneha	23.0	40.0	Ahmedabad

In [47]:
```
df.bfill()
```

Out[47]:

	Name	Age	Marks	City
0	Ajay	21.0	45.0	Rajkot
1	Neha	22.0	42.0	Morbi
2	Jatin	22.0	40.0	Ahmedabad
3	Sneha	23.0	40.0	Ahmedabad

In [ ]:
```
interpolate()
```

In [49]:
```
s = pd.Series([10,20,np.nan, 40, 50])
```

In [52]:
```
s
```

Out[52]:
```
0    10.0
1    20.0
2     NaN
3    40.0
4    50.0
dtype: float64
```

In [56]:
```
s = s.interpolate()
```

In [58]:
```
s
```

Out[58]:
```
0    10.0
1    20.0
2    30.0
3    40.0
4    50.0
dtype: float64
```

In [ ]:
```
nan ->
nat ->
None
```

In [60]:
```
data = {"Name" : ['Ajay', 'Neha', 'Jatin', 'Sneha'],
        "Dept" : ["CS","IT","CS","IT"],
        "Marks" : [25,26,24,30]}
```

In [62]:
```
df = pd.DataFrame(data)
```

In [64]:
```
df
```

Out[64]:

	Name	Dept	Marks
0	Ajay	CS	25
1	Neha	IT	26
2	Jatin	CS	24
3	Sneha	IT	30

In [68]:
```
df[df["Dept"] == "IT"]["Marks"].mean()
```

Out[68]:
```
np.float64(28.0)
```

In [78]:
```
df.groupby("Dept")[["Marks"]].max()
```

Out[78]:

Dept	Marks
CS	25
IT	30

In [80]:
```
df
```

Out[80]:

	Name	Dept	Marks
0	Ajay	CS	25
1	Neha	IT	26
2	Jatin	CS	24
3	Sneha	IT	30

In [84]:
```
df.groupby("Dept").size()  #counting record (rows) in each group
```

Out[84]:
```
Dept
CS    2
IT    2
dtype: int64
```

In [86]:
```
df = pd.DataFrame({"Dept" : ["CS","IT","IT","CS"],
                   "Marks" : [25, None, 30, 26]})
```

In [88]:
```
df
```

Out[88]:

	Dept	Marks
0	CS	25.0
1	IT	NaN
2	IT	30.0
3	CS	26.0

In [92]:
```
df["Dept"].count()
```

Out[92]:
```
np.int64(4)
```

In [94]:
```
df.groupby("Dept").size()
```

Out[94]:
```
Dept
CS    2
IT    2
dtype: int64
```

In [96]:
```
df.groupby("Dept")["Marks"].count()
```

Out[96]:
```
Dept
CS    2
IT    1
Name: Marks, dtype: int64
```

In [102…
```
df.groupby("Dept")["Marks"].agg(["max","min","mean"])
```

Out[102...

Dept	max	min	mean
CS	26.0	25.0	25.5
IT	30.0	30.0	30.0

In [126...

```
df.groupby("Dept").agg(
    Maximum_Marks=("Marks", "max")
)
```

Out[126...

Dept	Maximum_Marks
CS	25
IT	30

In [128...

```
df.groupby("Dept").agg(Maximum_Marks = ("Marks","max"))
```

Out[128...

Dept	Maximum_Marks
CS	25
IT	30

In [110...

```
df
```

Out[110...

	Dept	Marks
0	CS	25.0
1	IT	NaN
2	IT	30.0
3	CS	26.0

In [112...

```
data
```

Out[112...

```
{'Name': ['Ajay', 'Neha', 'Jatin', 'Sneha'],
 'Dept': ['CS', 'IT', 'CS', 'IT'],
 'Marks': [25, 26, 24, 30]}
```

In [114...

```
data["Semester"] = [1,1,3,3]
```

In [116...

```
data
```

Out[116...

```
{'Name': ['Ajay', 'Neha', 'Jatin', 'Sneha'],
 'Dept': ['CS', 'IT', 'CS', 'IT'],
 'Marks': [25, 26, 24, 30],
 'Semester': [1, 1, 3, 3]}
```

In [118…
```
df = pd.DataFrame(data)
```

In [124…
```
df.groupby(["Dept","Semester"])[["Marks"]].max()
```

Out[124…

		Marks
Dept	**Semester**	
CS	**1**	25
	3	24
IT	**1**	26
	3	30

In [35]:
```
import pandas as pd
import numpy as np
```

In [37]:
```
data = {'Name': ['Ajay', 'Neha', 'Jatin', 'Sneha'],
 'Dept': ['CS', 'IT', 'CS', 'IT'],
 'Marks': [25, 26, 24, 30]}
```

In [39]:
```
df = pd.DataFrame(data)
```

In [41]:
```
df
```

Out[41]:

	Name	Dept	Marks
0	Ajay	CS	25
1	Neha	IT	26
2	Jatin	CS	24
3	Sneha	IT	30

In [49]:
```
for i,j in df.groupby("Dept"):
    print(type(i))
    print(type(j))
```

```
<class 'str'>
<class 'pandas.core.frame.DataFrame'>
<class 'str'>
<class 'pandas.core.frame.DataFrame'>
```

In [61]:
```
df.groupby("Dept")["Marks"].agg(['mean','max','min'])
```

Out[61]:

	mean	max	min
Dept			
CS	24.5	25	24
IT	28.0	30	26

In [65]:
```
df.groupby("Dept").agg(Average_Marks = ("Marks","mean"), Highest_Marks = ("Marks
```

Out[65]:

Dept	Average_Marks	Highest_Marks
CS	24.5	25
IT	28.0	30

concat()

In [68]:
```
# pd.concat(objs, axis = 0, join = "outer", ignore_index = False)
```

In [70]:
```
batch1 = {"RollNo":[101,102,103,104,105],
          "Name" : ["Astha","Ansi","Dvija","Vidhi","Pritam"],
         "Marks" : [25,24,26,22,26]}
```

In [72]:
```
df1 = pd.DataFrame(batch1)
```

In [74]:
```
df1
```

Out[74]:

	RollNo	Name	Marks
0	101	Astha	25
1	102	Ansi	24
2	103	Dvija	26
3	104	Vidhi	22
4	105	Pritam	26

In [76]:
```
batch2 = {"RollNo":[106,107],
          "Name" : ["Bansi","Leena",],
         "Marks" : [25,24]}
```

In [78]:
```
df2 = pd.DataFrame(batch2)
```

In [80]:
```
df2
```

Out[80]:

	RollNo	Name	Marks
0	106	Bansi	25
1	107	Leena	24

In [88]:
```
final = pd.concat([df1,df2], axis = 0, ignore_index=True)
```

In [94]:
```
final.set_index("RollNo", inplace = True)
```

In [96]:
```
final
```

Out[96]:

RollNo	Name	Marks
101	Astha	25
102	Ansi	24
103	Dvija	26
104	Vidhi	22
105	Pritam	26
106	Bansi	25
107	Leena	24

In [98]:
```
df1 = pd.DataFrame({"Name" : ["Jay"], "Age" : [25]})
```

In [100…
```
df1
```

Out[100…

	Name	Age
0	Jay	25

In [102…
```
df2 = pd.DataFrame({"Name" : ["Smita"], "City" : ["Ahmedabad"]})
```

In [104…
```
df2
```

Out[104…

	Name	City
0	Smita	Ahmedabad

In [110…
```
pd.concat([df1,df2], join = "inner")
```

Out[110…

	Name
0	Jay
0	Smita

In [ ]:
```

```